TCJastrowFitting

December 2025

To-Do List

- Format Alignment.
  - ✓ Set up the To-Do list.
  - ☐ Set up the format of the raw data.
- Depth-Adjusted ee term investigation.
  - ✓ Alpha interpolation between varmin and emin freely-optimized jastrow (ano-vdz/cc-pvdz/cc-pCVDZ/aug-cc-pvqz)
  - ✓ Freely-optimized varmin AE results with ee cutoff is set to be 6. (aug-ano-v{d/t}z)
  - ✓ Standard freely-optimized emin results of AE (ano-vdz/cc-pCVDZ/aug-ano-v{d/t}z/aug-cc-pvqz)
  - ✓ Standard freely-optimized varmin results of AE (ano-vdz/cc-pCVDZ/aug-ano-v{d/t}z/aug-cc-pvqz).
  - ✓ Ee-depth-fixed AE calculation (ano-vdz, aug-ano-vd/tz; Depth: 0.0, -0.1, -0.2, -0.3, -0.4, -0.5).
  - ✓ AE calculation with the ee depth fixed at the fitting-optimal depth. (aug-ano-vdz: Depth: -0.33,-0.37)
  - ☐ ECP: Fit quadratic functions for total energies and do the same panel plots like for AE.
  - ☐ ECP and AE: Plot difference of total energy of fixed depth and total energy of free floating depth such that we can compare the total energies of AE and ECP in a fair manner.
  - ☐ Plot the free floating e-e depths of AE Jastrows like Kristoffer did for ECPs.
  - ☐ Plot out the max deviation with MAE, RMSE.
- Depth-Adjusted en/een term investigation.
  - ☐ Scale the depth/height of en/een term in Jastrow.

- Spin-dependent Jastrow.

  ✓ Standard freely-optimized varmin results of AE (Basis: aug-ano-v{d/t}z).

  ✓ Opposite-spin ee-depth-fixed AE calculation (Basis: aug-ano-vd/tz; Depth: 0.0, -0.1, -0.2, -0.3, -0.4, -0.5).

  ☐ Ee-cusp spin-dependent Jastrow.

Results

0.1 Free floating depths with varmin optimization (spin-independent)

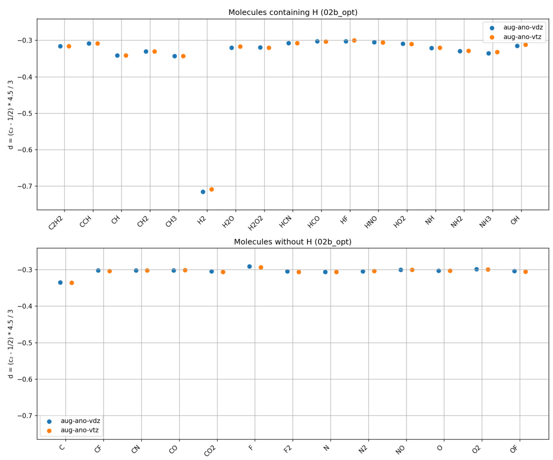

Figure 1: Free floating depths for AE calculations

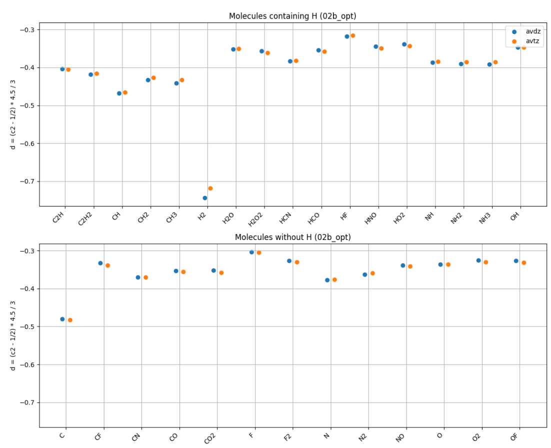

Figure 2: Free floating depths for ECP calculations

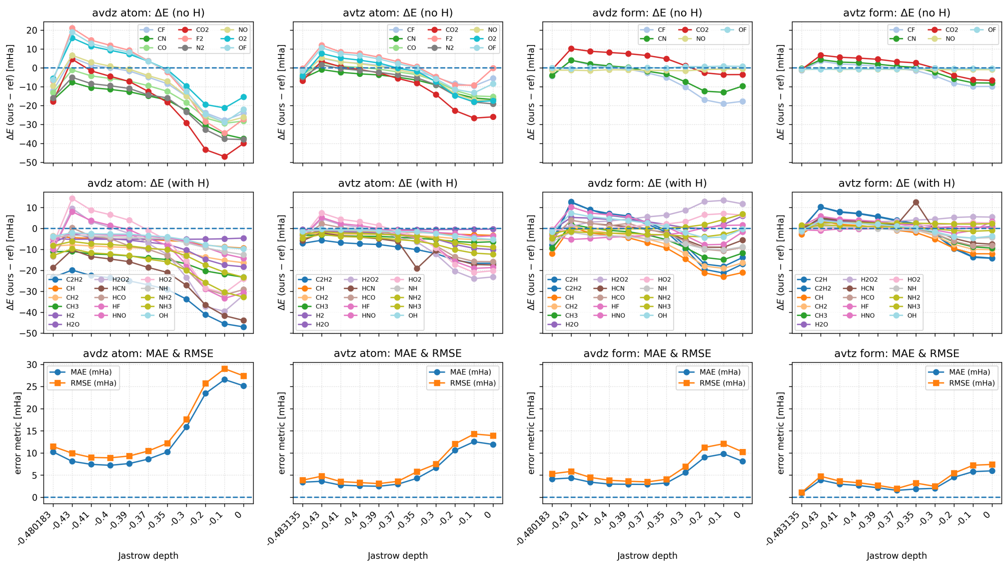

Figure 3: eCEPP

0.2 CCSD(T) energies against heat

0.2.1 eCEPP

0.2.2 ccECP

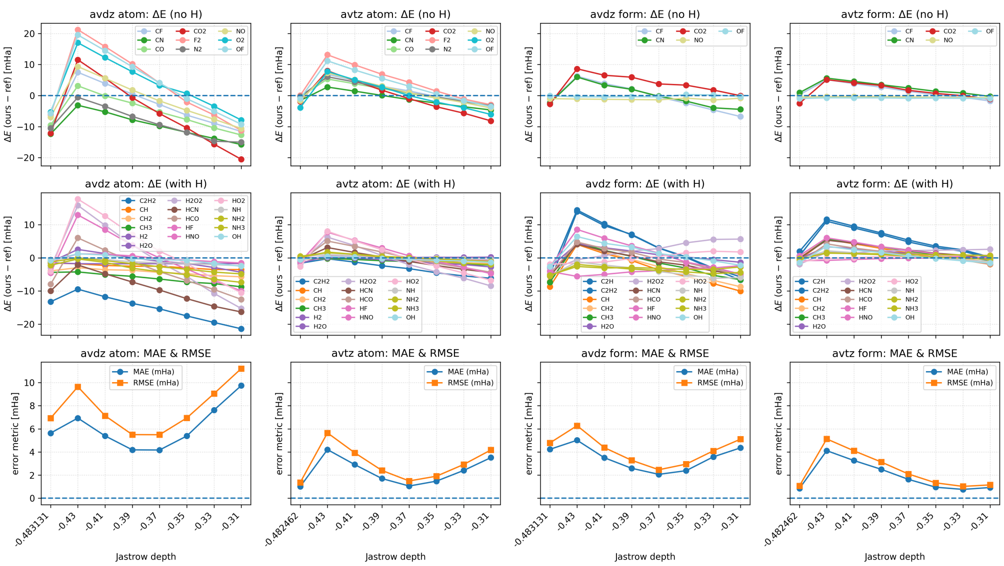

Figure 4: cECP

0.3 Depth patterns

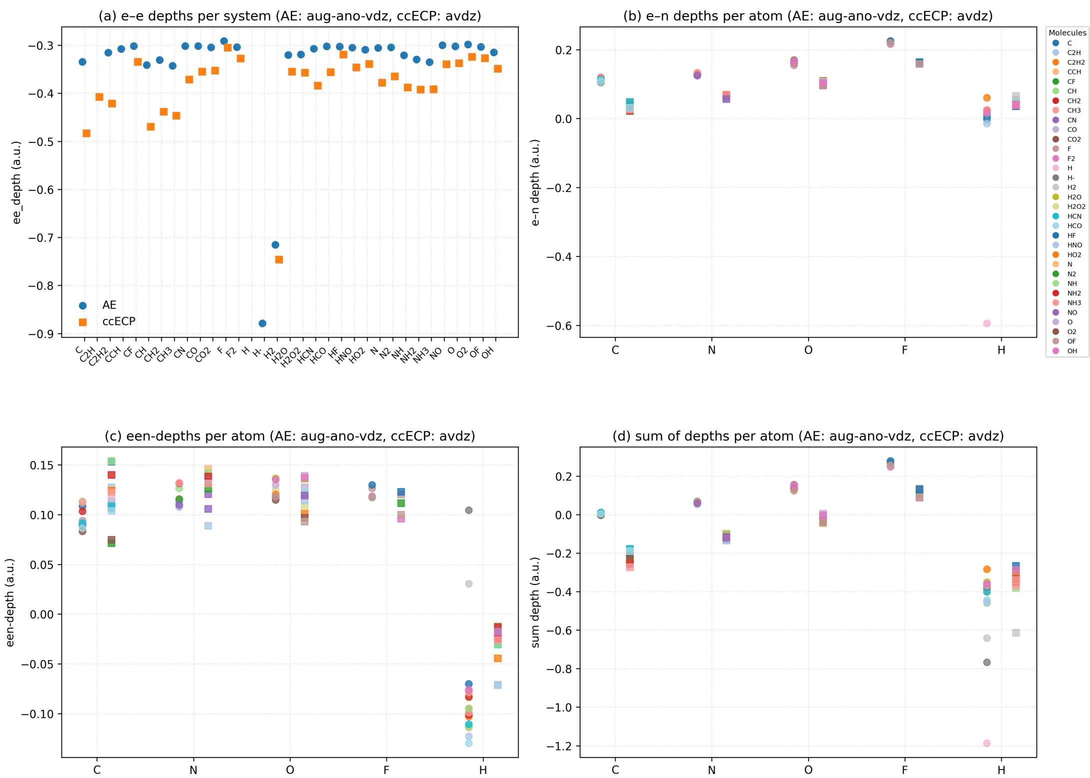

Figure 5: Patterns of free floating depths from varmin optimizations

1 Introduction